

Coffee Sales Performance Analysis

Excel Project Report

1. Project Overview

This project analyzes customer shopping behavior using transactional coffee sales data. The primary objective is to uncover key insights into spending patterns, regional sales trends, popular coffee types, and the impact of loyalty programs. This entire analysis is built as an end-to-end interactive dashboard project, fully developed utilizing Microsoft Excel's data processing and visualization capabilities.

2. Dataset Summary

The analysis relies on three primary data files tracking the sales ecosystem:

- **Customers:** Customer Name, Email, Country, and Loyalty Card Status.
- **Products:** Coffee Type (Arabica, Robusta, Liberica, Excelsa), Roast Type (Light, Medium, Dark), Package Size, and Unit Price.
- **Orders:** Order ID, Order Date, Customer ID, Product ID, and Quantity.

3. Data Cleaning & Preparation (in Excel)

Data preparation and transformation steps carried out within Excel to ensure data integrity:

- **Data Integration (Lookups):** Utilized XLOOKUP and INDEX/MATCH functions to merge customer details and product pricing into the central 'Orders' table, creating a robust master dataset.
- **Feature Engineering:** Generated a calculated Total Sales column by applying the formula $\text{=[Quantity]} * \text{[Unit Price]}$ across all rows.
- **Data Consistency:** Standardized date formats for timeline analysis, resolved missing values, and ensured appropriate data types for accurate Pivot Table aggregation.

4. Data Analysis using Pivot Tables

Structured analysis was performed using multiple Pivot Tables to answer strategic business questions:

Analysis Focus	Description & Insight Goal
Sales by Country	Compared total revenue across regions, mapping sales distribution primarily among the United States, United Kingdom, and Ireland.
Top 10 Customers	Identified the highest-spending customers based on total purchase amount to understand high-value segments and reward opportunities.
Sales Trend Over Time	Plotted monthly and yearly sales performance to visualize growth trajectories and seasonal demand spikes.
Loyalty Card Impact	Analyzed average spend and total revenue, directly comparing customers with active loyalty cards versus non-members.
Coffee & Roast Types	Conducted cross-tab analysis to determine which product combinations (e.g., Arabica Medium Roast vs. Robusta Dark Roast) generate the most volume.

5. Excel Interactive Dashboard

Finally, the insights generated from the Pivot Tables were consolidated into a dynamic, interactive Excel Dashboard to present the findings visually.

- **Key Metric Cards:** High-level highlights displaying Total Sales, Number of Orders, and Average Purchase Amount.
- **Visuals Used:** Bar charts for country and top customer comparisons, line charts for trend analysis, and Doughnut charts for loyalty program distribution.
- **Interactivity:** Integrated Slicers (by Roast Type, Coffee Type, Country) and Timeline Slicers (by Order Date) to allow stakeholders to interact dynamically with the data and visuals without altering the underlying formulas.

